

# SELECTION, NOT PERCEPTION: LEARNING TO CHOOSE DETECTOR CANDIDATES FOR ROBUST SHEET-IMAGE SCORE FOLLOWING

Pouya Mohseni, Ichiro Fujinaga

Schulich School of Music, McGill University, Montréal, Canada

## ABSTRACT

Real-time score following on sheet-music images needs no symbolic score, but systems trained on synthesized audio lose accuracy on real recordings. We show that for a state-of-the-art detection-based tracker this loss is a failure of selection rather than perception: on real room-microphone piano recordings, a correct candidate is among the detector’s top 256 boxes at every scored onset, and a causal oracle over them tracks 99.6% of onsets within 0.5 s, while the detector’s argmax tracks 80.0%. We therefore freeze the detector and learn to select. A parameter-free transition prior and a 23k-parameter set-based candidate scorer, trained only on synthesized audio, raise accuracy on the real recordings from 80.0% to 94.9%, above a recent end-to-end system that also needs layout annotations, with significant gains on 80 held-out pieces under noise. Teacher forcing shows that two thirds of the remaining error is error propagation, which history-independent candidate features reduce.

**Index Terms**— Score following, sheet music, real-time alignment, candidate re-ranking, music information retrieval

## 1. INTRODUCTION

Score following, the real-time alignment of a live performance to its score, underlies automatic page turning [1] and computer accompaniment [2, 3]. Most music exists only as printed or scanned sheets, and optical music recognition remains error-prone [4], so a line of work follows the performance directly on the sheet image, mapping audio and a score page to the current position [5, 6, 7, 8, 9, 10, 11].

These systems are trained on audio synthesized from the score, and real recordings remain their main open problem. On the public real piano performances aligned to sheet images [7], the detection-based CYOLO-SB [10] drops from 89.3% of onsets tracked within 0.5 s on synthetic audio to 79.9%, and CUNet from 85.5% to 22.4% [11]. Proposed remedies change the training: impulse-response augmentation [9], extra proprietary recordings [10], or a new cascaded architecture that also reads system and bar layout annotations at test time [11].

We start from a measurement instead. The CYOLO-SB detector proposes 8,112 candidate note boxes per step and the tracker reports the most confident one. On the real room-

microphone recordings, a box within the 0.5 s tolerance is among the 256 most confident candidates at every one of the 4,149 scored onsets, and an oracle that knows the answer but must commit one step at a time and may never move backwards tracks 99.6% of them. The argmax tracks 80.0% (Fig. 1). The information needed to follow the performance survives the room; what fails is the rule that reads a position out of it. For this model, robustness to real audio is a selection problem.

Our contributions are: (i) a diagnosis, by candidate-set oracles, that locates the synthetic-to-real gap of a sheet-image tracker in selection; (ii) a lightweight decoder for the frozen detector, a parameter-free transition prior and a 23k-parameter set-based scorer (1.6% of the detector) with neighbourhood features, which raises accuracy on the real recordings from 80.0% to 94.9% without layout annotations or retraining; (iii) an analysis showing that two thirds of the remaining error is error propagation, reduced by the features that do not depend on the tracker’s history.

**Related work.** Symbolic score followers infer the position with hidden (semi-)Markov models [12, 13] or online time warping [14]. Sheet-image trackers moved from staff strips [5] and reinforcement-learning agents [6, 7] to full-page segmentation [8] and audio-conditioned object detection [9, 10] with FiLM [15]. CODA [11] adds a system, bar and note cascade with learned transition penalties and beam search, trained end to end with layout supervision. We keep the detector and re-rank its output, akin to learned re-scoring of detections [16, 17]. Because the scorer conditions on its own past choices, it faces the exposure bias of sequence prediction [18, 19], which Sec. 3.5 measures.

## 2. METHOD

### 2.1. Candidates from a frozen detector

A performance is processed in steps  $t = 1, \dots, T$ . At each step the detector receives the audio up to  $t$  and the current page and outputs note boxes  $b_i = (x_i, y_i, w_i, h_i)$  with objectness  $o_i$  on three anchors of a  $52 \times 52$  grid. We keep the  $K = 256$  most confident,  $C_t$ . Staves are concatenated along one horizontal axis, the unrolled coordinate  $u_i$ , so that positions on either side

■ detector argmax (2.3 s off)     ground truth  
■ ours (0.05 s off)                      — top-40 candidates



**Fig. 1.** One onset of a real room-microphone recording. Grey: the detector’s 40 most confident note candidates. The argmax (red) sits on the wrong staff; the decoder (teal) picks a lower-ranked box on the true note (ring). The correct box was proposed but not chosen, the typical failure on real audio.

of a line break are close. The baseline outputs

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} o_i. \quad (1)$$

Everything below reads only  $\mathcal{C}_t$  and the decoder’s own previous choice  $\hat{u}_{t-1} = u_{i_{t-1}}$ .

## 2.2. Transition prior

Between steps  $\Delta_t$  frames apart, the performer moves forward by a roughly predictable amount. We score the displacement  $d_i = u_i - \hat{u}_{t-1}$  with a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left( -\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

where  $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ ,  $\mu = 6$  px per reference step of  $\Delta_0 = 5$  frames,  $\sigma = 18$  px, and the floor  $J = -6$  keeps jumps possible. The prior has no trained parameters (**constants: provenance to be stated after the held-out re-selection**). Alone, the hand score  $h_i = \log o_i + \log \pi_i$  already improves the tracker (Table 1).

## 2.3. Candidate scorer

Each candidate is described by  $\phi_i \in \mathbb{R}^{33}$ : confidence (log objectness, gap to the frame’s best, rank); box geometry; motion relative to the history (normalized displacement, backward and staff-change flags, residual to a constant-velocity extrapolation); relation to the detector’s predicted system and bar boxes; and a *neighbourhood* group asking whether the box looks like a real notehead (candidates within 25 and 100 px, gaps to the nearest candidate on each side, rank in reading order, size and aspect relative to the frame’s median box, position inside the predicted bar). Only the confidence, geometry and neighbourhood groups are independent of the history. We also read the 128-dimensional feature  $f_i$  from which the detector’s head

scores the box. A permutation-invariant network [20] embeds the set and pools a context shared by all candidates:

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[ \frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

where  $g$  and  $r$  are two-layer perceptrons (64, 32 and 64, 1 units),  $W \in \mathbb{R}^{64 \times 128}$ , and the maximum is per dimension, so each candidate is judged against the others in its frame. The scorer has 22,881 parameters, 1.6% of the 1.46M-parameter detector.

## 2.4. Training and decoding

We run the frozen detector over the MSMD training pieces [21], synthesized and convolved with measured room impulse responses [22], and store  $\mathcal{C}_t$  and  $f_i$  at every onset. With  $\varepsilon_i$  the error in frames of candidate  $i$ , the soft target is  $q_i \propto \exp(-\varepsilon_i/\tau)$  with  $\tau = 3$ , and the loss is the listwise cross-entropy

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_{j \in \mathcal{C}_t} \exp s_j}. \quad (5)$$

History features come from the best candidates of the previous steps (teacher forcing), perturbed on 30% of samples by Laplace noise of scale 30 px. We train with AdamW (learning rate  $3 \cdot 10^{-3}$ , cosine schedule, 40 epochs, batch 256) and keep the epoch with the best greedy rollout on the validation pieces. At test time the learned and hand scores are blended,

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

with  $\beta = 0.7$  (**provenance as for the prior**), and  $\hat{u}_t$  becomes the history of step  $t + 1$ . Decoding is causal and greedy: one forward pass of the scorer over 256 candidates per step.

# 3. EXPERIMENTS

## 3.1. Setup

**Data.** The detector is the released CYOLO-SB checkpoint [10]. The scorer is trained on its candidates over the 353 MSMD training pieces and validated on the 19 validation pieces. The test set is the real-performance subset of [7] (Setting II in [11]): 16 pieces, 25 pages and 4,149 onsets played on a Yamaha AvantGrand N2 hybrid piano and recorded with a room microphone. A larger held-out set uses 80 pieces withheld from the scorer’s training; as the detector saw their synthesized audio, we render them with a real room impulse response and additive noise at 12, 6, 3 and 0.5 dB SNR (40,594 onsets per level).

**Metric and protocol.** As in [10, 11], we report the ratio of onsets whose predicted position maps to within 0.05 to 5 s of the true time, with 0.5 s as the primary figure, and 95%

intervals from a bootstrap over pieces [23]. The standard protocol steps a tracker only at annotated onsets; since our decoder carries state, we also step it on every frame at 20 fps and score the onset frames, which removes access to annotated onset timing. Retraining with another seed moves the test score by about a point, so we report seed means, and the single reported checkpoint is chosen without the test set: the best held-out configuration mean, then its best seed on held-out.

### 3.2. Main result

Table 1 compares the decoder with published systems on the same real recordings. Our run of the released detector reproduces the published CYOLO-SB figures to within 0.001 at every threshold. The transition prior alone, without training, adds 6.5 points at 0.5 s, and the scorer another 8.4. The decoded tracker is more accurate than every published system at every threshold, including CYOLO-SB+A, which uses extra training data, and CODA, which retraining the whole model and reads layout annotations at test time. Against the detector’s argmax, the gain at 0.5 s is 13.8 points under the piece bootstrap (95% CI [5.6, 23.0]). Stepped on every frame, the decoder scores **X** at 0.5 s (**every-frame run pending; the argmax is unchanged at .800 by construction**).

### 3.3. Ablation

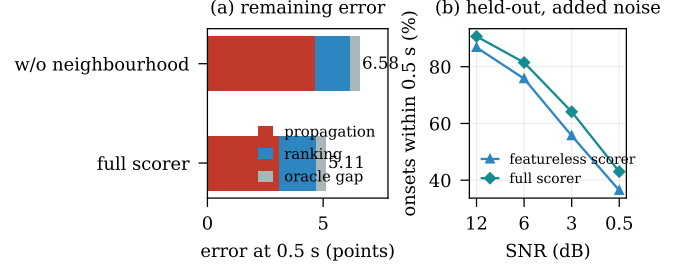
Table 2 adds the decoder’s parts one at a time, and each helps on the real recordings. The prior’s gain is significant on its own (95% CI [2.7, 10.0]), and so is the scorer’s gain over the prior ([1.4, 7.6]). The per-candidate features raise the real-recording mean from 91.4 to between 92.7 and 93.2; with 16 pieces this step is not separable from zero there, but on the 80 held-out pieces all six seeds beat the featureless scorer with intervals excluding zero (mean gain 3.2 points). The two projection widths are tied within seed noise on both sets; we report the wider one because it has the better held-out mean, the rule fixed before reading the test set.

### 3.4. Robustness to noise

On the held-out pieces (Fig. 2b) the full scorer improves on the featureless one by 3.9, 5.7, 8.5 and 6.6 points at 12, 6, 3 and 0.5 dB, each with a 95% interval excluding zero. As a share of the remaining error the gain shrinks from 30% to 10%: the candidate features help most where the detector’s boxes are still reliable.

### 3.5. What is left

Given the true previous position at every step (teacher forcing), the same scorer reaches 98.0%, and the causal oracle 99.6%. The deployed tracker’s 5.1-point error thus splits into 3.1 points of error propagation, 1.6 of ranking, and 0.4 no causal selector could recover (Fig. 2a). The scorer without the



**Fig. 2.** (a) Error on the real recordings at 0.5 s, split by teacher forcing into error propagation (lost by conditioning on the tracker’s own past choices), ranking (lost even with the true history) and the gap to a causal oracle over the same candidates. (b) Held-out pieces with a real room impulse response and additive noise (40,594 onsets per level).

neighbourhood features (Table 2, fourth row; one checkpoint each) splits 4.7 and 1.5: those features cut propagation by a third and leave ranking unchanged. This matches what they measure. Displacement and velocity features are computed from the previous position and mislead the scorer once it is wrong, whereas whether a box looks like a real notehead does not depend on the history. Recovering from a lost state directly is harder: the chosen box’s objectness does reveal that the tracker is lost (AUC 0.81), but lost steps are about 5% of all steps, so no threshold reaches a precision above 0.33, and gated re-localization lowered accuracy in every configuration we tried. Most remaining errors are timing drift within two bars (190 of 212); jumps to a wrong staff fall from 90 with the featureless scorer to 19.

## 4. CONCLUSION

A state-of-the-art sheet-image tracker loses accuracy on real recordings mostly because of how it selects a position, not what it perceives, and a small decoder over its frozen candidates recovers most of the gap, reaching 94.9% of onsets within 0.5 s on real piano recordings. As in prior work, the correct page is given and the audio restarts at page turns; all performances are piano; and with 16 test pieces, differences of a point or two between variants are not resolvable, hence the seed means and the larger held-out set. The remaining error is propagation, and what reduces it is evidence independent of the tracker’s history; agreement between the pitches a candidate implies and those heard in the audio is a natural next source. Code and trained scorers will be released.

**Table 1.** Real piano recordings (16 pieces, 4,149 onsets, room microphone). Ratio of onsets tracked within each error threshold. Published rows are from Table 1 of [11], which also lists weaker systems [5, 7, 8]; <sup>†</sup> trained with additional proprietary recordings. *Layout*: system and bar annotations needed at test time. Our rows use the released CYOLO-SB detector, frozen; the scorer is the checkpoint selected on held-out pieces (over six seeds it scores  $0.927 \pm 0.015$  at 0.5 s).

| Method                                     | Onset error threshold (s) |             |             |             |             | Layout |
|--------------------------------------------|---------------------------|-------------|-------------|-------------|-------------|--------|
|                                            | $\leq 0.05$               | $\leq 0.1$  | $\leq 0.5$  | $\leq 1.0$  | $\leq 5.0$  |        |
| CYOLO [9]                                  | .563                      | .581        | .712        | .749        | .919        | no     |
| CYOLO-SB [10]                              | .610                      | .630        | .799        | .832        | .960        | no     |
| CYOLO-SB+A <sup>†</sup> [10]               | .682                      | .706        | .865        | .891        | .981        | no     |
| CODA [11]                                  | .721                      | .743        | .883        | .914        | .987        | yes    |
| CYOLO-SB, our run (argmax, Eq. 1)          | .609                      | .629        | .800        | .832        | .960        | no     |
| + transition prior (Eq. 2), 0 params       | .636                      | .660        | .865        | .899        | .973        | no     |
| + candidate scorer (ours), 23k params      | <b>.755</b>               | <b>.777</b> | <b>.949</b> | <b>.975</b> | <b>.994</b> | no     |
| + candidate scorer, stepped on every frame | .                         | .           | .           | .           | .           | no     |

**Table 2.** Decoder ablation on the frozen detector. *Real*: accuracy at 0.5 s on the real recordings, mean  $\pm$  s.d. over seeds (count in brackets). *Held-out*: 80 unseen pieces at 12 dB SNR, seed mean.

| Decoder                  | Params | Real               | Held-out |
|--------------------------|--------|--------------------|----------|
| Argmax (1)               | 0      | 80.0               | .        |
| + prior (2)              | 0      | 86.5               | .        |
| + scorer, base features  | 9,697  | 91.4 [1]           | 86.8     |
| + motion, detector feat. | 11,497 | 92.2 $\pm$ 1.0 [5] | 89.3     |
| + neighbourhood feat.    | 12,073 | 93.2 $\pm$ 0.7 [8] | 89.7     |
| wider projection (ours)  | 22,881 | 92.7 $\pm$ 1.5 [6] | 90.0     |

## 5. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

## 6. ACKNOWLEDGMENTS

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